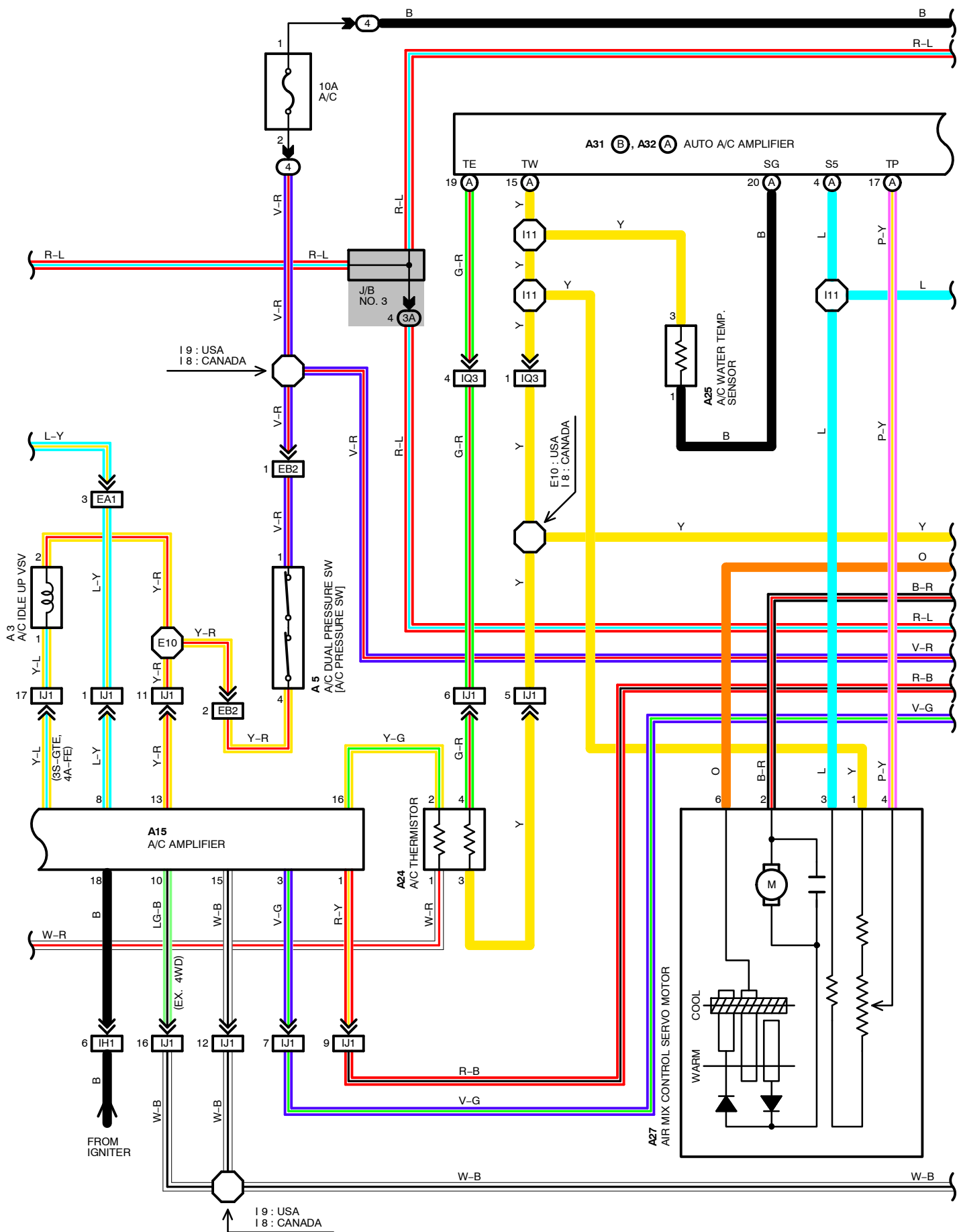
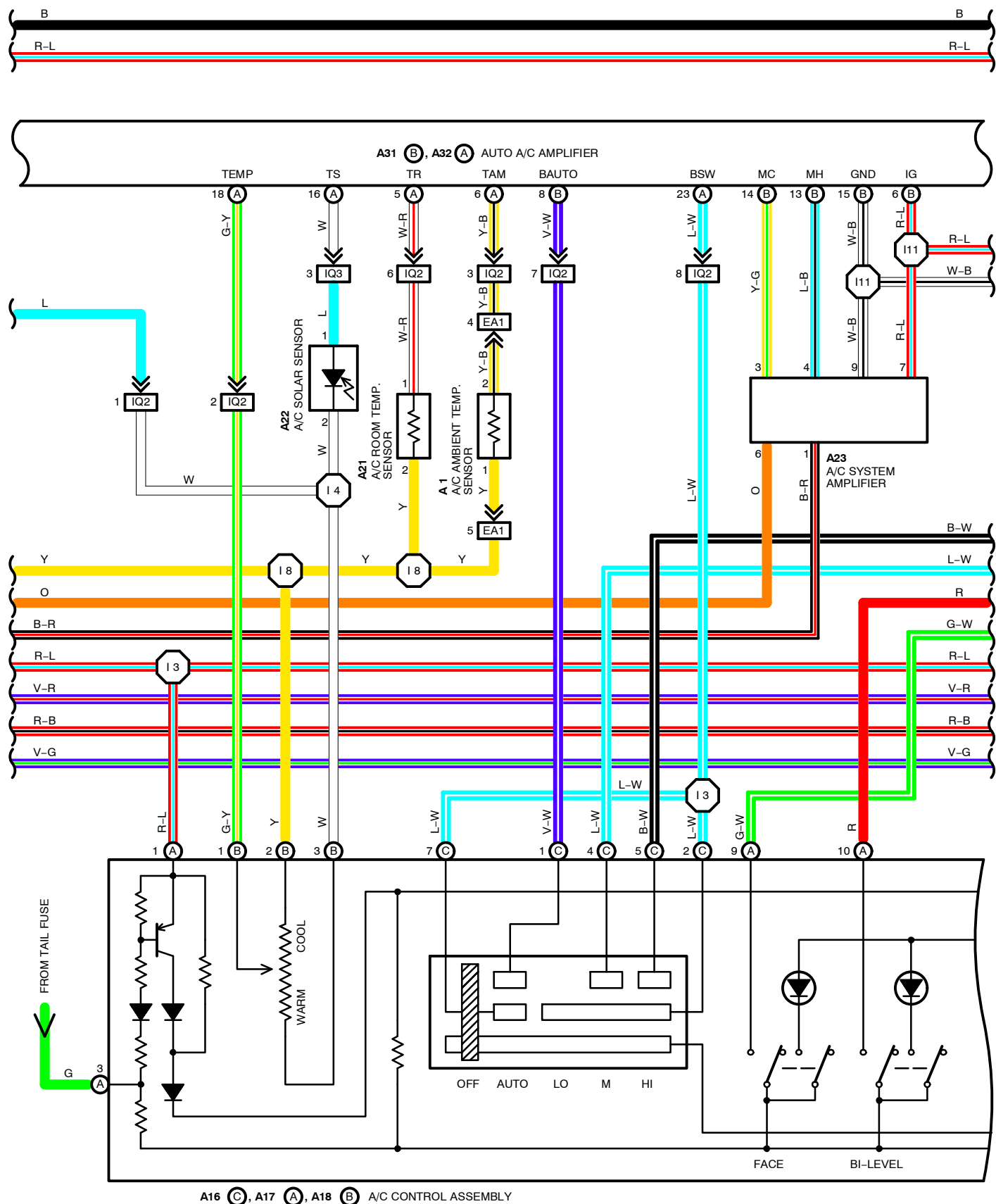
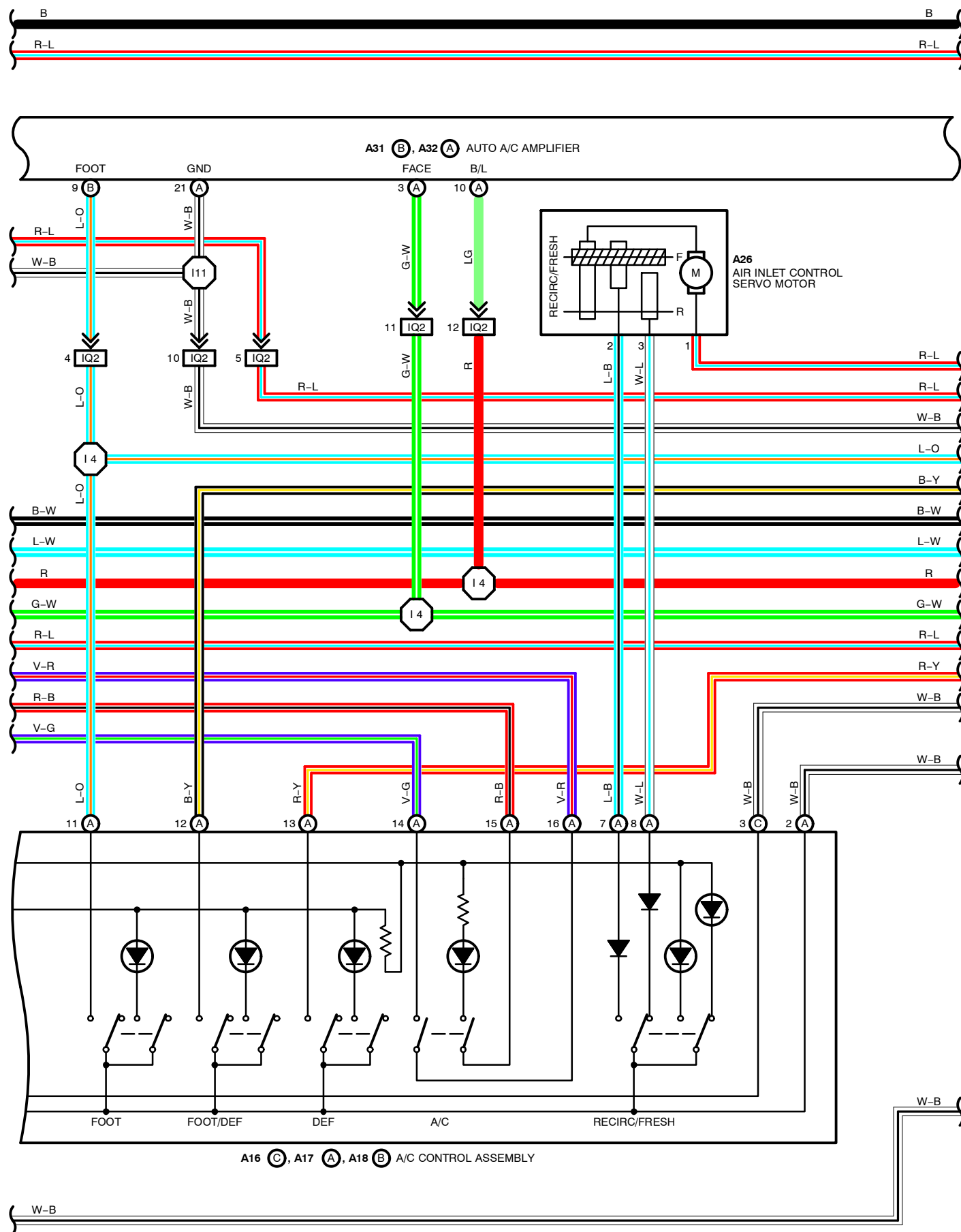






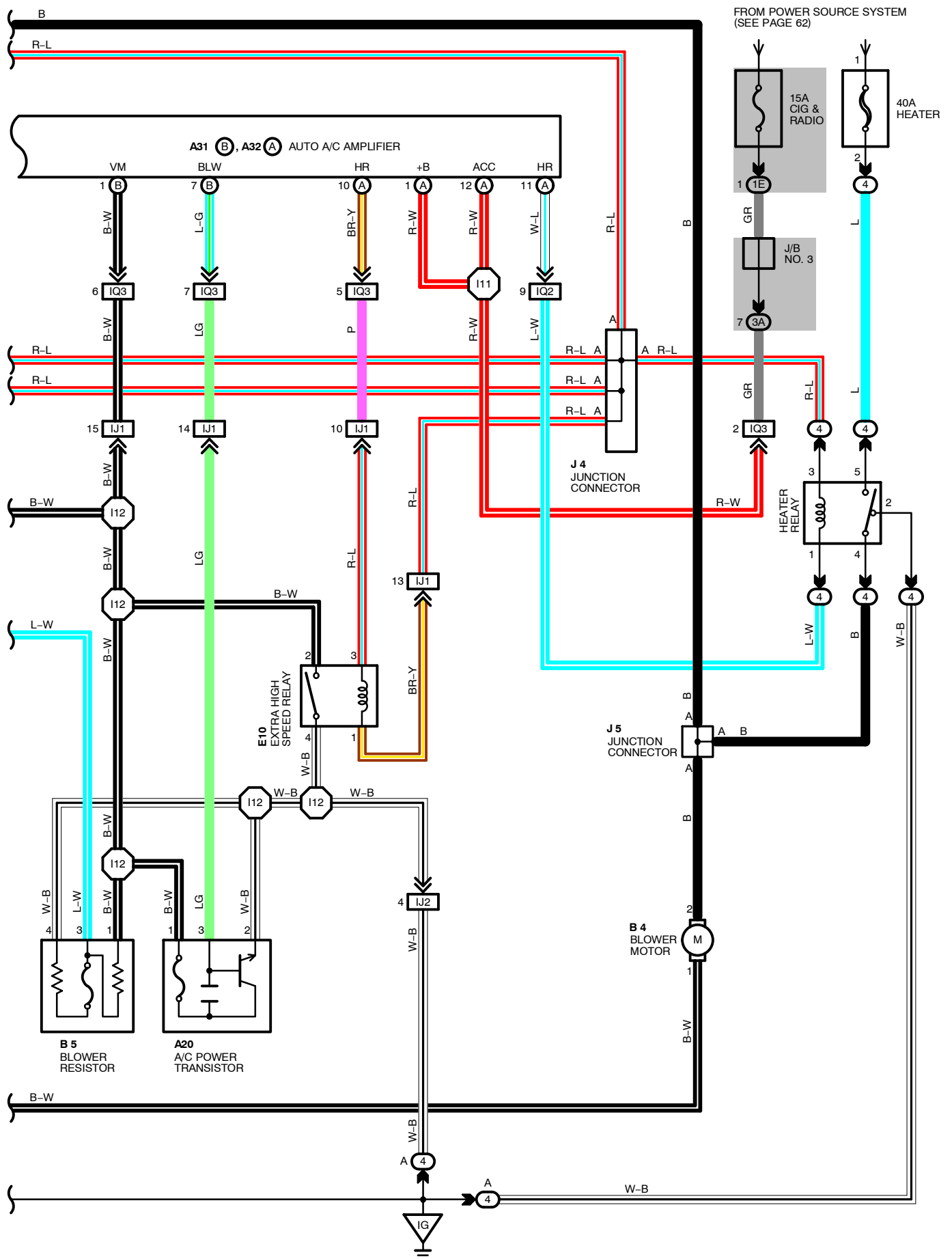
RADIATOR FAN AND AIR CONDITIONING (AUTOMATIC AIR CONDITIONING)





A31 (B), A32 (A) AUTO A/C AMPLIFIER





RADIATOR FAN AND AIR CONDITIONING (AUTOMATIC AIR CONDITIONING)

SYSTEM OUTLINE

1. COOLING FAN OPERATION

WHEN THE IGNITION SW IS TURNED ON, THE CURRENT FROM IGN FUSE FLOWS TO **TERMINAL 1** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 2** → **TERMINAL 3** OF THE A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 2** OF WATER TEMP. SW (FOR FANS CONTROL) → **TERMINAL 1** → **GROUND**, FROM **TERMINAL 1** OF A/C FAN RELAY NO. 2 → **TERMINAL 2** → **TERMINAL 3** OF A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 2** OF WATER TEMP. SW (FOR FANS CONTROL) → **TERMINAL 1** → **GROUND**, CAUSING RELAY NO. 1 AND RELAY NO. 2 OF EACH FAN TO TURN ON.

* OPERATION AT LOW SPEED

WHEN THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON AND THE A/C OPERATES, THE CURRENT FLOWS FROM GAUGE FUSE FLOWS TO **TERMINAL 1** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 2** → **TERMINAL 8** OF A/C AMPLIFIER CAUSING A/C MAGNETIC CLUTCH RELAY TO TURN ON.

AT THAT TIME, THE CURRENT FROM GAUGE FUSE FLOWS TO **TERMINAL 5** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 3** → **TERMINAL 4** OF A/C MAGNETIC CLUTCH → **GROUND**, AND FROM **TERMINAL 3** OF A/C MAGNETIC CLUTCH RELAY **TERMINAL 1** OF A/C FAN RELAY NO. 3 → **TERMINAL 2** → **GROUND**.

AS A RESULT, A/C MAGNETIC CLUTCH AND A/C FAN RELAY NO. 3 TURNS ON AND THE CURRENT FLOWS FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 3** OF A/C FAN RELAY NO. 2 → **TERMINAL 5** → **TERMINAL 2** OF RADIATOR FAN MOTOR → **TERMINAL 1** → **GROUND**, FLOWING TO EACH FAN MOTOR IN SERIES, CAUSING THE COOLING FAN TO ROTATE AT LOW SPEED.

* OPERATION AT HIGH SPEED

DURING A/C OPERATION, WHEN THE PRESSURE OF A/C COMPRESSOR BECOMES HIGHER THAN NORMAL PRESSURE (MORE THAN 14.3 KG/CM² 1401 KPA, 203 PSI), THE A/C PRESSURE SW TURNS OFF.

AS A RESULT, FAN RELAY NO. 1 AND NO. 2 TURNS OFF AND THE CURRENT FLOWS FROM RDI FAN FUSE TO → **TERMINAL 4** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 3** → **TERMINAL 2** OF RADIATOR FAN MOTOR → **TERMINAL 1** → **GROUND**, AND FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 1** OF A/C FAN RELAY NO. 2 → **TERMINAL 3** → **GROUND**, FLOWING TO EACH FAN MOTOR IN PARALLEL CAUSING THE COOLING FAN TO ROTATE AT HIGH SPEED. WHEN THE ENGINE COOLANT TEMPERATURE BECOMES MORE THAN ABOUT 90° (194°F), THE WATER TEMP. SW TURNS OFF AND THE SAME OPERATION AS ABOVE IS PERFORMED.

2. HEATER BLOWER MOTOR OPERATION (DIAL TYPE BLOWER CONTROL SW (W/ AUTO A/C))

CURRENT IS APPLIED AT ALL TIMES THROUGH THE HEATER FUSE TO **TERMINAL 5** OF HEATER RELAY. WHEN THE IGNITION SW IS TURNED ON, CURRENT FLOWS THROUGH GAUGE FUSE TO **TERMINAL 3** OF HEATER RELAY → **TERMINAL 1** → **TERMINAL HR** OF AUTO A/C AMPLIFIER. AT THE SAME TIME, CURRENT ALSO FLOWS FROM GAUGE FUSE TO **TERMINAL IG** OF AUTO A/C AMPLIFIER, AND **TERMINAL (A) 1** OF A/C CONTROL ASSEMBLY.

* LOW SPEED OPERATION (OPERATION AT MANUAL)

CURRENT TO **TERMINAL 3** OF HEATER RELAY FLOWING TO **TERMINAL 1** OF HEATER RELAY → **TERMINAL HR** → **TERMINAL BSW** → **TERMINAL (C) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND** AND TURNS THE HEATER RELAY ON. THIS CAUSES THE CURRENT TO **TERMINAL 5** OF THE HEATER RELAY TO FLOW TO **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 4** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT LOW SPEED.

* HIGH SPEED OPERATION (OPERATION AT MANUAL)

CURRENT TO **TERMINAL 3** OF HEATER RELAY FLOWS TO **TERMINAL 1** → **TERMINAL HR** OF THE AUTO A/C AMPLIFIER → **TERMINAL BSW** → **TERMINAL (C) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND**, TURNING THE HEATER RELAY ON.

AT THE SAME TIME, CURRENT FLOW TO **TERMINAL 5** OF THE HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL (C) 5** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT HIGH SPEED.

* MEDIUM SPEED OPERATION (OPERATION AT MANUAL M1, M2)

THE CURRENT TO **TERMINAL 3** OF HEATER RELAY → **TERMINAL 1** → **TERMINAL HR** OF THE A/C AMPLIFIER → **TERMINAL BSW** → **TERMINAL (C) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND**, AND TURNS THE HEATER RELAY ON.

AS A RESULT, THE CURRENT TO **TERMINAL 5** OF HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 3** → **TERMINAL (C) 4** OF A/C CONTROL ASSEMBLY → **TERMINAL (C) 3** → **GROUND**, AND CAUSES THE BLOWER MOTOR TO ROTATE AT MEDIUM SPEED.

* AUTO FUNCTION

WHEN THE AUTO SW (A/C CONTROL ASSEMBLY) IS TURNED ON, SIGNALS ARE INPUT TO **TERMINAL BSW** AND **TERMINAL BAUTO** OF AUTO A/C AMPLIFIER CONTROLLING THE CURRENT FLOW FROM **TERMINAL BLW** OF THE AUTO A/C AMPLIFIER TO **TERMINAL 3** OF POWER TRANSISTOR → **TERMINAL 2** → **GROUND**, THE AUTO A/C AMPLIFIER CONTROLS THE BLOWER MOTOR STEPLESSLY.

3. OPERATION OF AIR INLET CONTROL SERVO MOTOR

(SWITCHING FROM FRESH TO RECIRC)

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 1** OF AIR INLET SERVO MOTOR. WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE RECIRC SIDE, TO CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 2** → **TERMINAL (A) 7** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**. THE MOTOR ROTATES AND THE DAMPER MOVES TO THE RECIRC SIDE.

WHEN IT IS IN THE **RECIRC** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

(SWITCHING FROM RECIRC TO FRESH)

WITH IGNITION SW ON, WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE FRESH SIDE, THE CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 3** → **TERMINAL (A) 8** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**, THE MOTOR ROTATES AND THE DAMPER MOVES TO THE FRESH SIDE. WHEN IT IS IN THE **FRESH** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

4. OPERATION OF AIR VENT MODE CONTROL SERVO MOTOR

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 5** OF AIR VENT MODE CONTROL SERVO MOTOR → **TERMINAL 6** → **GROUND**, AND THE DAMPER MOVES TO THE POSITION OF THE MODE SELECTION SW OF THE CONTROL ASSEMBLY SWITCH IS ON. WHEN THE MODE SELECTION SW OF A/C CONTROL ASSEMBLY IS MOVED TO **DEF** POSITION FROM THE DAMPER IN THE **FACE** POSITION, THE CURRENT FLOWS FROM **TERMINAL 7** OF AIR VENT MODE CONTROL SERVO MOTOR TO **TERMINAL (A) 13** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**. AS A RESULT, THE SERVO MOTOR OPERATES UNTIL THE DAMPER REACHES **DEF** POSITION.

WHEN THIS OCCURS THE CURRENT TO THE A/C CONTROL ASSEMBLY IS SHUT OFF AND ROTATION OF THE MOTOR STOPS. SWITCHING TO OTHER MODES IS CONTROLLED BY THE SERVO MOTOR ACCORDING THE FLOWING CURRENT:

1. **FOOT/DEF** POSITION: THE CURRENT FLOWS FROM **TERMINAL 4** OF SERVO MOTOR TO **TERMINAL (A) 12** OF A/C CONTROL ASSEMBLY.
2. **FOOT** POSITION: THE CURRENT FLOWS FROM **TERMINAL 3** OF SERVO MOTOR TO **TERMINAL (A) 11** OF A/C CONTROL ASSEMBLY.
3. **BI-LEVEL** POSITION: THE CURRENT FLOWS FROM **TERMINAL 2** OF SERVO MOTOR TO **TERMINAL (A) 10** OF A/C CONTROL ASSEMBLY.

5. OPERATION OF AIR MIX CONTROL SERVO MOTOR

WHEN THE TEMPERATURE CONTROL VOLUME IS TURNED TO THE COOL SIDE, A SIGNAL IS INPUT TO **TERMINAL TEST** OF AUTO A/C FAMPLIFIER FROM **TERMINAL (B) 1** OF A/C CONTROL ASSEMBLY.

AS A RESULT, A SIGNAL IS OUTPUT FROM **TERMINAL AMH** OF AUTO A/C AMPLIFIER TO **TERMINAL 3** OF A/C SYSTEM AMPLIFIER AND THE CURRENT FLOWING TO **TERMINAL 7** OF A/C SYSTEM AMPLIFIER FROM THE GAUGE FUSE FLOWS FROM **TERMINAL 1** OF A/C SYSTEM AMPLIFIER → **TERMINAL 2** OF AIR MIX CONTROL SERVO MOTOR → **TERMINAL 6** → IF /AC SYSTEM AMPLIFIER → **TERMINAL 9** → **GROUND**. CAUSING THE AIR MIX CONTROL SERVO MOTOR TO ROTATE.

AT THIS TIME IS INPUT THE DAMPER OPENING ANGLE FROM **TERMINAL 4** OF SERVO MOTOR TO **TERMINAL TP** OF AUTO A/C AMPLIFIER THIS IS USED TO DETERMIN THE DAMPER **STOP** POSITION AND MAINTAIN THE SET TEMPERATURE.

WHEN THE TEMPERATURE CONTROL VOLUME IS TURNED TO THE WARM SIDE, IN A/C SYSTEM AMPLIFIER THE CURRENT FLOWS FROM SERVO MOTOR → **TERMINAL 6** OF A/C SYSTEM AMPLIFIER → **TERMINAL 6** OF AIR MIX CONTROL SERVO MOTOR → MOTOR → **TERMINAL 2** → **TERMINAL 1** OF A/C SYSTEM AMPLIFIER → **TERMINAL 9** → **GROUND**, ROTATING THE MOTOR IN REVERSE AND SWITCHING THE DAMPER FROM COOL TO WARM SIDE.

RADIATOR FAN AND AIR CONDITIONING (AUTOMATIC AIR CONDITIONING)

SYSTEM OUTLINE

6. AIR CONDITIONING OPERATION

THE A/C AMPLIFIER RECEIVES VARIOUS SIGNALS, THE ENGINE RPM SIGNAL FROM THE IGNITER, OUTLET TEMPERATURE SIGNAL FROM THE A/C THERMISTOR, COOLANT TEMPERATURE FROM THE ENGINE COOLANT TEMP. SENSOR (EFI WATER TEMP. SENSOR), AND THE LOCK SIGNAL FROM THE A/C COMPRESSOR, ETC.

WHEN THE ENGINE IS STARTED AND THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON, THE CURRENT FLOWS FROM A/C FUSE TO **TERMINAL 3** OF A/C AMPLIFIER.

AS A RESULT, THE CURRENT FLOWING FROM **TERMINAL 2** OF A/C MAGNETIC CLUTCH RELAY TO **TERMINAL 8** OF A/C AMPLIFIER FLOWS FROM **TERMINAL 15** OF A/C AMPLIFIER TO **GROUND** AND TURNS IN THE MAGNETIC CLUTCH RELAY.

BECAUSE THE MAGNETIC CLUTCH IS ON, THE A/C COMPRESSOR OPERATES, CAUSING THE CURRENT FLOWING FROM A/C IDLE-UP VSV TO **TERMINAL 6** OF A/C AMPLIFIER TO FLOW TO **TERMINAL 15** OF A/C AMPLIFIER → **GROUND**, AND TURNS ON THE VSV TO AVOID LOWERING THE ENGINE RPM DURING AIR CONDITIONER OPERATION.

WHEN ANY OF THE FOLLOWING SIGNALS ARE INPUT TO THE A/C AMPLIFIER, THE AMPLIFIER OPERATES TO TURN OFF THE A/C:

- * ENGINE HIGH RPM SIGNAL
- * COOLANT TEMP. SIGNAL IS HIGH.
- * A SIGNAL THAT THE TEMPERATURE AT THE AIR OUTLET IS LOW.
- * A SIGNAL THAT THERE IS A LARGE DIFFERENCE BETWEEN ENGINE RPM AND COMPRESSOR RPM
- * A SIGNAL THAT THE REFRIGERANT PRESSURE IS ABNORMALLY HIGH OR LOW.

SERVICE HINTS

A 4 A/C MAGNETIC CLUTCH

4-GROUND : APPROX. 3.7 Ω

A 5 A/C PRESSURE SW

3-2 : OPEN ABOVE APPROX. 13.5 KG/CM² (192 PSI, 1323 KPA)

CLOSED BELOW APPROX. 10 KG/CM² (142 PSI, 980 KPA)

1-4 : OPEN WITH PRESSURE LESS THAN 2.1 KG/CM² (30 PSI, 206 KPA) OR ABOVE 27 KG/CM² (384 PSI, 2648 KPA)

A15 A/C AMPLIFIER

8-15 : CONTINUITY WITH A/C SW (A/C CONTROL ASSEMBLY) ON AND IGNITION SW **ON** POSITION

14-15 : ALWAYS CONTINUITY

14-GROUND: ALWAYS CONTINUITY

15-GROUND: ALWAYS CONTINUITY

13-GROUND: APPROX. 12 VOLTS WITH IGNITION SW ON

A21 A/C ROOM TEMP. SENSOR

1-2 : APPROX. 1.7 KΩ AT 25°C (77°F)

A24 A/C THERMISTOR

1-2, 3-4 : APPROX. 4852 Ω AT 0°C (59°F)

APPROX. 2341 Ω AT 15°C (59°F)

APPROX. 1500 Ω AT 25°C (77°F)

A27 AIR MIX CONTROL SERVO MOTOR

2-GROUND : APPROX. 12 VOLTS WITH TEMPERATURE CONTROL VOLUME AT **WARM** TO **COOL** POSITION

6-GROUND : APPROX. 12 VOLTS WITH TEMPERATURE CONTROL VOLUME AT **COOL** TO **WARM** POSITION

1-3 : ALWAYS APPROX. 6 KΩ

A31(B), A32(A) AUTO A/C AMPLIFIER

B-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ACC** OR **ON** POSITION

IG-GROUND : APPROX. 12 VOLTS WITH IGNITION SW **ON** POSITION

HR-GROUND : CONTINUITY WITH BLOWER SW (A/C CONTROL ASSEMBLY) ON OR AUTO SW (A/C CONTROL ASSEMBLY) ON

S5-GROUND : APPROX. 5 VOLTS WITH IGNITION SW ON

SG-GROUND : ALWAYS CONTINUITY

REC-GROUND: APPROX. 12 VOLTS WITH A/C CONTROL ASSEMBLY AT **RECIRC** POSITION

FRS-GROUND: APPROX. 12 VOLTS WITH A/C CONTROL ASSEMBLY AT **FRESH** POSITION

GND-GROUND: ALWAYS CONTINUITY

B 5 BLOWER RESISTOR

1-3 : APPROX. 0.84 Ω

3-4 : APPROX. 1.48 Ω

W 3 WATER TEMP. SW (FOR FANS CONTROL)

1-2 : OPEN ABOVE APPROX. 90°C (194°F)

CLOSED BELOW APPROX. 83°C (181.4°F)

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A 1	28 (3S-GTE), 30 (5S-FE)	A21	32	B 4	32
A 2	28 (3S-GTE), 30 (5S-FE)	A22	32	B 5	32
A 3	28 (3S-GTE), 30 (5S-FE)	A23	32	C 1	A 28 (3S-GTE), 30 (5S-FE)
A 4	28 (3S-GTE), 30 (5S-FE)	A24	32	C 2	B 28 (3S-GTE), 30 (5S-FE)
A 5	28 (3S-GTE), 30 (5S-FE)	A25	32	E10	33
A15	32	A26	32	J 4	33
A16	C 32	A27	32	J 5	33
A17	A 32	A28	32	J 6	33
A18	B 32	A31	B 32	R 1	28 (3S-GTE), 30 (5S-FE)
A20	32	A32	A 32	W 3	28 (3S-GTE), 30 (5S-FE)

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)
5	27	R/B NO. 5 (ENGINE COMPARTMENT FRONT RIGHT)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2D	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2E		
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	36 (3S-GTE) 38 (5S-FE)	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
EB2	36 (3S-GTE) 38 (5S-FE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
IG1	42	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE CONTROL MODULE (ECU))
IG2		
IH1	42	ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ1	44	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ2		
IQ2	44	COWL WIRE AND A/C NO. 2 WIRE (BESIDE HEATER UNIT)
IQ3		

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (5S-FE)	FRONT RIGHT FENDER
EB	38 (5S-FE)	FRONT LEFT FENDER
ID	42	LEFT KICK PANEL
IG	42	R/B NO. 4 SET BOLT

○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 2	36 (3S-GTE) 38 (5S-FE)	ENGINE ROOM MAIN WIRE	E10	38 (5S-FE)	ENGINE WIRE
E 3	36 (3S-GTE) 38 (5S-FE)		E13	38	ENGINE ROOM MAIN WIRE
E 4	36 (3S-GTE) 38 (5S-FE)		I 3	44	COWL WIRE
E 5	36 (3S-GTE) 38 (5S-FE)		I 4		
E 7	36 (3S-GTE) 38 (5S-FE)		I 8		
E10	36 (3S-GTE)		I 9		
			I11	44	A/C NO. 2 WIRE
			I12	44	A/C NO. 1 WIRE

RADIATOR FAN AND AIR CONDITIONING (AUTOMATIC AIR CONDITIONING)

